APPENDIX A
PROOFS

A. Proof of Proposition 1

Proof: TREC applies the recurrence in the main paper to each candidate p_j at t_i, considering only predecessors p_k with $d_{\mathcal{M}}(p_k, p_j) \le s(t_i - t_{i-1})$. Keeping the highest-scoring prefix at each endpoint preserves the optimal substructure [1]. Backtracking through the stored predecessors from an endpoint maximizing ω_{nj} recovers a feasible trajectory with maximum coverage.

For complexity, the nested loops over n timestamps and all $|\mathcal{P}|^2$ candidate pairs yield $O(n \cdot |\mathcal{P}|^2)$. Each distance check $d_{\mathcal{M}}(p_h, p_j)$ can be performed in $O(1)$ time after a standard preprocessing of candidate-pair distances, a routine step also adopted in related studies [2], [3]. Thus the dynamic-programming phase of TREC takes $O(n \cdot |\mathcal{P}|^2)$ time. ∎

B. Proof of Lemma 1

Proof: Each group begins with one candidate, so its diameter is zero. Before adding p_j, TRRP checks both $d_{\mathcal{M}}(p_j, p_k) < \delta$ and $d_{\mathcal{M}}(p_k, p_j) < \delta$ for every existing member p_k. This matters because the road network is directed. Every ordered pair in the enlarged group then has distance below δ, and induction gives the stated bound for the completed group. ∎

C. Proof of Proposition 2

Proof: Each density score uses a coordinate range query followed by at most ν network-distance checks. A group query returns at most ν candidates, and checking each against at most ν current members costs $O(\nu^2)$ for that group. At most $|\mathcal{P}|$ groups are formed, giving $O(|\mathcal{P}|(\log|\mathcal{P}| + \nu^2))$ for grouping. Choosing representatives scans the candidates once. TREC then takes $O(nr^2)$ on the r representatives, which yields the stated bound. ∎

D. Proof of Lemma 2

Proof: Since $\mathcal{P}$ is finite and distinct candidates are distinct positions on $\mathcal{M}$ with $d_{\mathcal{M}}(p, q) > 0$, both maxima range over finite sets, so L is a well-defined finite constant. The bound then holds by construction: for any timestamp t_i and any $p_j \neq p_k$ in $\mathcal{P}$, the ratio $|\Omega_i(p_j) - \Omega_i(p_k)|/d_{\mathcal{M}}(p_j, p_k)$ appears in the maximum (in one of the two orders), hence is at most L; for $p_j = p_k$ both sides vanish. Intuitively, L is controlled by the crossing density of coverage-region boundaries at the candidate granularity: the coverage count changes only where a path crosses such a boundary, so L is governed by the density of base stations, the geometry of their coverage regions, and the sampling resolution of $\mathcal{P}$, and is independent of the timestamp and of the particular pair chosen. ∎

E. Proof of Proposition 3

Proof: Let $P^* = \{p_1^*, \ldots, p_n^*\}$ denote the optimal trajectory over $\mathcal{P}$, and for each timestamp i let $r_{\theta_i} \in \mathcal{R}$ denote the representative selected from the group θ_i containing p_i^* (here r_{θ_i} denotes a representative point, not the count $r = |\mathcal{R}|$). Since p_i^* and r_{θ_i} belong to the same group, Lemma 1 gives $d_{\mathcal{M}}(p_i^*, r_{\theta_i}) < \delta$, and Lemma 2 (both points are candidates) yields the per-timestamp replacement loss

$$\Omega_i(p_i^*) - \Omega_i(r_{\theta_i}) \le L \cdot d_{\mathcal{M}}(p_i^*, r_{\theta_i}) \le L\delta. \tag{1}$$

The projected sequence $\hat{P} = \{r_{\theta_1}, \ldots, r_{\theta_n}\}$, however, is not necessarily feasible: by the triangle inequality, $d_{\mathcal{M}}(r_{\theta_i}, r_{\theta_{i+1}}) \le d_{\mathcal{M}}(p_i^*, p_{i+1}^*) + 2\delta$, so consecutive representatives may exceed the speed budget by up to 2δ. We therefore construct a feasible surrogate trajectory $Q = \{q_1, \ldots, q_n\} \subseteq \mathcal{R}$ inductively: set $q_1 = r_{\theta_1}$, and obtain q_{i+1} from q_i as follows. Follow a shortest path in $\mathcal{M}$ from q_i toward $r_{\theta_{i+1}}$, truncating it at arc length $s \cdot (t_{i+1} - t_i)$ if it is longer; retreat the resulting endpoint by $\delta + g$ along the path, where g is the candidate granularity, and snap the retreated point to its nearest representative w; set $q_{i+1} = w$. Since candidates are sampled at granularity g, the retreated point lies within g of some candidate, whose group representative is within δ; hence $d_{\mathcal{M}}(q_i, w) \le s \cdot (t_{i+1} - t_i) - (\delta + g) + (\delta + g) = s \cdot (t_{i+1} - t_i)$, so reachability is preserved at every step. Because every candidate lies within distance δ of its representative, each step increases the residual deviation $d_{\mathcal{M}}(q_i, r_{\theta_i})$ by at most $c\delta$ for a constant c: at most 3δ is lost when approaching $r_{\theta_{i+1}}$ along a shortest path within the speed budget, at most 2δ when snapping to the nearest representative, and the retreat adds a further $\delta + g$, so relative to the naive construction c grows by $O(g/\delta)$; the form of the bound is unaffected and only γ is adjusted accordingly. Hence $d_{\mathcal{M}}(q_i, r_{\theta_i}) \le c(i-1)\delta$. Combining both losses via Lemma 2 and the triangle inequality, and setting $\gamma := c/2$, we obtain

$$\begin{aligned}
\Omega(P^*) - \Omega(Q) &\le \sum_{i=1}^{n} \Big[\Omega_i(p_i^*) - \Omega_i(r_{\theta_i}) \\
&\qquad + \Omega_i(r_{\theta_i}) - \Omega_i(q_i) \Big] \qquad (2) \\
&\le \sum_{i=1}^{n} \left(\delta + c(i-1)\delta\right) L \qquad (3) \\
&\le (2n + \gamma n^2) L\delta. \qquad (4)
\end{aligned}$$

When the remaining path length after q_i is shorter than $\delta + g$, the retreat is truncated at q_i itself and the snap target is q_i's own group representative, so every proxy step remains feasible; the $O(g/\delta)$ term in c covers the residual snap drift in all cases.

Finally, Q is a feasible trajectory over the representative set $\mathcal{R}$, while TRRP returns the optimal trajectory P' over $\mathcal{R}$ by invoking TREC (Proposition 1). Therefore $\Omega(P') \ge \Omega(Q) \ge \Omega(P^*) - (2n + \gamma n^2) L\delta$. ∎

F. Proof of Proposition 4

Proof: From Proposition 3, we have the coverage bound $\Omega(P') \geq \Omega(P^*) - (2n + \gamma n^2)L\delta$. Substituting the chosen grouping diameter $\delta = \frac{\epsilon\Omega_{\max}}{(2n+\gamma n^2)L}$ into the inequality yields $\Omega(P') \geq \Omega(P^*) - (2n+\gamma n^2)L \cdot \frac{\epsilon\Omega_{\max}}{(2n+\gamma n^2)L} = \Omega(P^*) - \epsilon\Omega_{\max}$. Since $\Omega(P^*) \geq \Omega_{\max}$, it follows that $\Omega(P') \geq \Omega(P^*) - \epsilon\Omega_{\max} \geq \Omega(P^*) - \epsilon\Omega(P^*) = (1-\epsilon)\Omega(P^*)$. Thus, TRRP achieves a $(1-\epsilon)$ approximation guarantee. ∎

G. Proof of Proposition 5

Proof: First, the shared candidate-pair distance table uses $O(|\mathcal{P}|^2)$ storage; its graph-dependent construction time is outside the per-trajectory bound. Once it is available, each reachability check takes $O(1)$. With k-step lookahead, identifying the maximum-coverage candidates and selecting the anchor $\bar{p}_{i+k}$ both take $O(|\mathcal{P}|)$. For each candidate at t_i, computing the score defined in the main paper requires $O(|\mathcal{P}|)$ in total, and selecting the best candidate is subsumed in this bound. Thus each iteration costs $O(|\mathcal{P}|)$, leading to $O(n|\mathcal{P}|)$ across all timestamps. The stated $O(n|\mathcal{P}|)$ recovery bound excludes that one-off table construction. ∎

H. Proof of Proposition 6

Proof: Let $\mathcal{F}_i(P'_{i-1})$ be the set of feasible candidates at timestamp t_i given the prefix $\{p'_1, \ldots, p'_{i-1}\}$ of the approximate solution, and similarly $\mathcal{F}_i(P^*_{i-1})$ for the optimal solution. To measure the similarity between the feasible sets, we define the coverage ratio λ_i for each timestamp t_i as $\lambda_i = \frac{\max_{p\in\mathcal{F}_i(P'_{i-1})}\Omega_i(p)}{\max_{q\in\mathcal{F}_i(P^*_{i-1})}\Omega_i(q)}$.

Since $p^*_i \in \mathcal{F}_i(P^*_{i-1})$, we have $\max_{q\in\mathcal{F}_i(P^*_{i-1})}\Omega_i(q) \geq \Omega_i(p^*_i)$, and thus $\max_{p\in\mathcal{F}_i(P'_{i-1})}\Omega_i(p) \geq \lambda_i \cdot \Omega_i(p^*_i)$. If the denominator of λ_i is zero we set $\lambda_i = 1$; then $\Omega_i(p^*_i) = 0$, so the timestamp's term $\lambda_i\,\Omega_i(p^*_i)$ vanishes and it drops out of the bound. The algorithm selects a point $p'_i \in \mathcal{F}_i(P'_{i-1})$ by maximizing the score function defined in the main paper, so $S_i(p'_i) \geq S_i(p)$ for any $p \in \mathcal{F}_i(P'_{i-1})$. In particular, let p be the point achieving the maximum coverage in $\mathcal{F}_i(P'_{i-1})$. By the $\min$ in the score definition in the main paper, the movement-consistency term $(1-\alpha)\cdot\big(1-\min\{1,\, d_{\mathcal{M}}(p,\bar{p}_{i+k})/D^{\max}_{i+k}\}\big)$ lies in $[0, 1-\alpha]$ for *every* candidate, regardless of whether the anchor $\bar{p}_{i+k}$ is reachable within the lookahead window; it may therefore be dropped from the right-hand side unconditionally, yielding:

$$\alpha\cdot\frac{\Omega_i(p'_i)}{C^{\max}} + (1-\alpha)\cdot\left(1-\min\left\{1,\frac{d_{\mathcal{M}}(p'_i,\bar{p}_{i+k})}{D^{\max}_{i+k}}\right\}\right) \geq \alpha\cdot\frac{\Omega_i(p)}{C^{\max}}. \tag{5}$$

Rearranging and combining with the earlier inequality for $\max_{p\in\mathcal{F}_i(P'_{i-1})}\Omega_i(p)$, we get

$$\Omega_i(p'_i) \geq \lambda_i\cdot\Omega_i(p^*_i) - \frac{(1-\alpha)}{\alpha}C^{\max}\cdot\left(1-\min\left\{1,\frac{d_{\mathcal{M}}(p'_i,\bar{p}_{i+k})}{D^{\max}_{i+k}}\right\}\right). \tag{6}$$

Each parenthesized term lies in $[0,1]$ by construction. For the last k timestamps no anchor is applied, so the score reduces to the coverage term and $\Omega_i(p'_i) \geq \lambda_i\cdot\Omega_i(p^*_i)$ holds directly without penalty. Summing over $i=1$ to n therefore yields

$$\begin{aligned}\Omega(P') &= \sum_{i=1}^{n}\Omega_i(p'_i) &(7)\\ &\geq \sum_{i=1}^{n}\lambda_i\cdot\Omega_i(p^*_i) - \frac{(1-\alpha)}{\alpha}\sum_{i=1}^{n-k}\Big[C^{\max}\cdot\left(1-\min\left\{1,\frac{d_{\mathcal{M}}(p'_i,\bar{p}_{i+k})}{D^{\max}_{i+k}}\right\}\right)\Big] &(8)\\ &\geq \Omega(P^*)\cdot\min_{i\in[1,n]}\lambda_i - \frac{(1-\alpha)}{\alpha}nC^{\max}, &(9)\end{aligned}$$

where $C^{\max} = \max_{i\in[1,n],\,p\in\mathcal{P}}\Omega_i(p)$ is the global maximum coverage. ∎

I. Proof of Proposition 7

Proof: Every retained candidate has the same coverage score in $\mathcal{M}$ and $\mathcal{M}'$. In particular, all candidates in the observed regions U remain available. Feasibility is preserved because for any consecutive points p_i, p_{i+1}, a shortest path π in $\mathcal{M}$ can be decomposed into segments either inside R or outside with endpoints on ∂R. Internal segments are included in $\mathcal{M}'$, and for external segments, shortest paths between boundary points are preserved. Replacing each segment by its preserved path yields a path in $\mathcal{M}'$ of the same length, so $d_{\mathcal{M}'}(p_i,p_{i+1}) = d_{\mathcal{M}}(p_i,p_{i+1})$ and the speed constraint is maintained.

For retained candidates, the preserved road distances leave the speed constraint $d_{\mathcal{M}'}(p_i,p_{i+1}) \leq s(t_{i+1}-t_i)$ unchanged. Together with the unchanged coverage scores, this gives the full-network optimum a feasible counterpart with the same coverage on $\mathcal{M}'$. Conversely, since $\mathcal{M}' \subseteq \mathcal{M}$, every subnetwork trajectory is feasible on the full network, so the subnetwork optimum cannot exceed the full-network optimum. Therefore the optimal coverage is preserved exactly: $\Omega(P') = \Omega(P^*)$.

The result concerns optimal coverage, not the precise point sequence chosen when several optima exist. TRRP and TRHA operate on the localized candidate set in the same way as on the full set. Their bounds apply with the coverage and distance parameters of that localized set. ∎

J. Proof of Proposition 8

Proof: The time complexity analysis follows from applying the TRSN optimization to each algorithm's base complexity. The key insight is that by partitioning the problem into independent subnetworks, we can analyze the complexity per subnetwork and then sum over all subnetworks.

For TREC, the base complexity is $O(n|\mathcal{P}|^2)$. After partitioning, for each subnetwork θ with M_θ cellular record sequences and candidate set size reduced to $\eta_\theta|\mathcal{P}|$ due to spatial

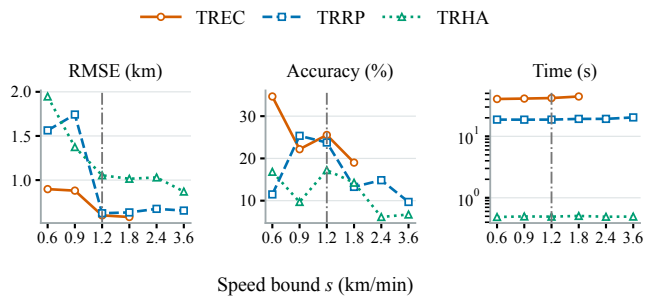


Fig. 1. Direct speed-bound sweep on GZ. The vertical line marks $s = 1.2$ km/min. TREC is reported for $s \leq 1.8$ km/min. Time is per trajectory on a logarithmic scale

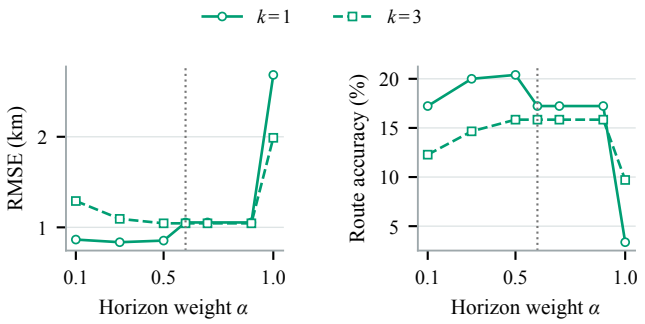


Fig. 2. Horizon-weight comparison on GZ for $k = 1$ and $k = 3$. The vertical line marks $\alpha = 0.6$

localization, the complexity becomes $O(M_\theta n \eta_\theta^2 |\mathcal{P}|^2)$. Summing over all subnetworks gives $O(\sum_{\theta=1}^{|\Theta|} M_\theta n \eta_\theta^2 |\mathcal{P}|^2)$. For TRRP, grouping in subnetwork θ takes $O(\eta_\theta |\mathcal{P}|(\log |\mathcal{P}| + \nu^2))$, including the pairwise checks within each local group. Recovery on its r_θ representatives takes $O(n_\theta r_\theta^2)$ per sequence. Summing over the M_θ sequences in each subnetwork gives the stated expression. The distance tables are computed once per subnetwork and excluded from the per-sequence costs. For TRHA, the base algorithm has complexity $O(n|\mathcal{P}|)$. After partitioning, the per subnetwork complexity is $O(n\eta_\theta|\mathcal{P}|)$ for M_θ sequences, giving total complexity $O(\sum_{\theta=1}^{|\Theta|} M_\theta n \eta_\theta |\mathcal{P}|)$.

The reduction ratios η_θ, $0 < \eta_\theta < 1$ capture the spatial localization effect, representing the fraction of the original candidate set that remains relevant within each subnetwork θ. ■

APPENDIX B
SUPPLEMENTARY EXPERIMENTS

A. Direct Speed-Bound Sweep

To test how the movement constraint helps resolve ambiguous cellular observations, we vary the speed bound s directly on GZ. Figure 1 complements the percentile-based sweep in the main paper by expressing s in km/min. RMSE and road-segment accuracy are aggregated over the evaluated positions, and runtime is averaged per trajectory.

Across the three methods, recovery quality generally improves as s increases, then declines when the bound becomes too permissive. Small bounds exclude plausible transitions, whereas overly large bounds admit more competing paths. These trends favor an intermediate value of s; the main paper describes its selection using percentiles of a speed distribution estimated from station-based positions.

B. Horizon Weight across Window Sizes

Future observations guide TRHA's local choices; this experiment tests how their contribution depends on the look-ahead window. On GZ, we sweep the horizon weight α for $k = 1$ and $k = 3$, as shown in Figure 2. Smaller weights give more influence to the future anchor, whereas larger weights favor current coverage.

Combining current and future evidence gives a better balance of error and route accuracy than using current coverage alone. The shorter window achieves its best recovery quality with moderate weights, while the longer window benefits from increasing the weight on current coverage before the anchor contribution disappears. These trends support weighting future guidance according to its horizon, so that the anchor helps distinguish locally plausible candidates.

C. Per-Edge Speed Limits

On GZ, Per-edge uses OpenStreetMap `maxspeed` tags. Missing values are estimated from connected or nearby roads of matching class, directionality, and lane count, followed by class- and lane-based statistics and road-class defaults. Per-edge speeds are capped at 45 km/h. A transition is feasible when $T_{\mathcal{M}}(p, q) \leq \Delta t$, where $T_{\mathcal{M}}$ is the shortest road-network travel time under these speeds and Δt is the elapsed time between records.

Table I shows that incorporating per-edge speed limits reduces RMSE and improves road-segment accuracy for all three methods. TRHA shows the largest reduction in positional error, while TREC and TRRP also benefit from the refined movement constraints.

TABLE I
CONSTANT AND PER-EDGE SPEED BOUNDS ON GZ

Method	Speed rule	RMSE (km)	Acc. (%)
TREC	Constant	0.534	35.8
	Per-edge	**0.529**	**37.2**
TRRP	Constant	0.529	35.0
	Per-edge	**0.514**	**35.8**
TRHA	Constant	0.822	28.1
	Per-edge	**0.742**	**28.9**

D. Preprocessing Cost and Reuse

Efficient recovery also depends on how much preparation can be reused. We therefore time the complete pipeline on GZ and compare the initial computation with subsequent recovery using cached distances. For the same inputs, Figure 3(a) separates input and candidate preparation, shared candidate-pair distance construction, algorithm-specific preparation, and recovery. TRRP's additional preparation includes grouping and construction of its representative-pair table. Candidate

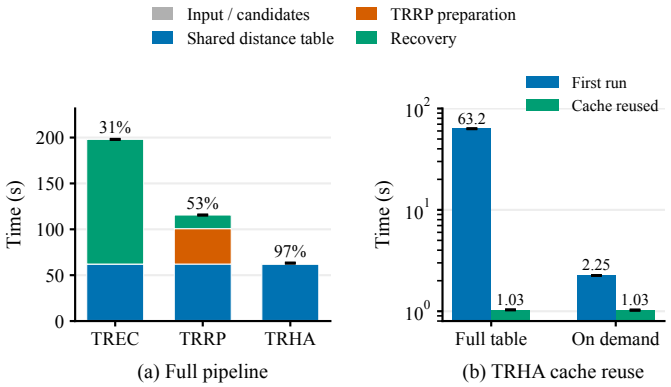


Fig. 3. Preprocessing and recovery costs on GZ: (a) complete pipeline, with labels showing the shared distance table's fraction of total time; (b) TRHA with full precomputation or on-demand caching, before and after cache reuse. Error bars show standard deviations; panel (b) uses a logarithmic axis

granularity is $g = 0.4$ km and group diameter is $\delta = 0.6$ km. Times are averaged over three fresh-cache runs, with the full shared construction cost included in each method's total.

Both approximation methods reduce total pipeline time relative to TREC, even when shared preprocessing is included. Recovery dominates TREC's cost; representative compression and sequential selection reduce that work in TRRP and TRHA, making shared distance construction a larger fraction of their first runs. Reusing distance tables and TRRP's representative structures allows subsequent recovery on the same inputs to avoid repeating that preparation.

For TRHA, Figure 3(b) further compares full precomputation with on-demand caching of exact single-source shortest-path distances. Dijkstra searches run on the original or reversed graph for source- or destination-based queries, respectively, and retain the resulting distance vectors. Computing only the queried vectors substantially reduces first-run time and distance-array storage. Once the caches are populated, the two modes have similar recovery times and return identical positions and coverage scores in every repetition. Thus, on-demand caching preserves the recovery result while reducing the initial preprocessing cost.

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